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Spatial patterns of tree seedling establishment and their relationship to environmental variables in a cold-temperate deciduous forest of eastern North America

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Abstract

Patterns of seedling recruitment may have persistent effects on population and community processes. Assuming seed availability is not limiting, the environmental sieve (i.e., the suite of factors influencing seed germination and seedling emergence and survival) determines how many seedlings establish and, most importantly, where they do so. In this study, we identify the spatial structure of some resources and abiotic conditions known to be significant for tree seedling emergence and survival and determine how these environmental factors influence the establishment of *Fagus grandifolia*, *Acer saccharum*, *Fraxinus americana*, and *Ostrya virginiana* in a deciduous forest of southern Québec (Canada). We expect an increase from *Fagus*, through *Acer* and *Fraxinus*, to *Ostrya* in the control of environmental variables on seedling emergence and survival, because of differences in the seed size of these species. Density of newly-emerged seedlings of all four species showed positive spatial autocorrelation at distances of up to *ca*. 10 m. Environmental variables were also structured at the same spatial scale, except for soil moisture. *Acer* seedling emergence pattern was positively correlated to photosynthetic photon flux density (PPFD), and the pattern of *Fraxinus* to soil N and moisture. Seedling survival was not spatially autocorrelated for any of the four species, although it was positively density-dependent in *Acer* and *Fagus*. In only *Ostrya* was seedling survival correlated (positively) to one of the environmental variables studied, i.e., PPFD. Overall, environmental variables were spatially less heterogeneous than seedling emergence and survival. Either seed availability was not saturating or factors not considered here, such as competition and predation (the intensity of which often varies with resources and/or abiotic conditions), modified the influence that the physical environment had on patterns of seedling establishment. Our prediction of a greater environmental control on seedling emergence and survival in small-seed species was not totally confirmed.

Introduction

Seedling recruitment represents a demographic bottleneck for most populations, particularly for those of species that do not have the ability to grow clonally (Gurevitch et al. 2002). Assuming seed availability is not limiting, the environmental sieve (i.e., the multitude of factors influencing seed germination and seedling emergence and survival) determines how many seedlings establish and survive and, most importantly, where they do so (Harper 1977; George and Bazzaz 1999; Nathan and Muller-Landau 2000).

Several studies in forest ecosystems have shown how abiotic factors (resources and conditions) vary in space, even at the scale of a few centimeters or a few meters (Lechowicz and Bell 1991; Molofsky and Augspurger 1992; Farley and Fitter 1999). Because different factors typically have different and discordant spatial patterns, there often is a high within-habitat spatial heterogeneity in microsite suitability for tree seedling establishment (Dwyer and Merriam 1981; Beatty and Stone 1986; Houle 1992a). Such microsite heterogeneity may contribute to increased local diversity assuming species have specialized in the regeneration niche hyperspace (Grubb 1977; Collins and Good 1987; Nakashizuka 2001; Beckage and Clark 2003).

Of the resources known to influence tree seedling emergence and survival, light has been well-studied (Canham and Marks 1985; Poulson and Platt 1989; Vàzquez-Yanes and Orozco-Segovia 1990; Nicotra et al. 1999). Typically, small-seed tree species are shade-intolerant and require more light to recruit seedlings (e.g., recruitment limited to recent gaps or large-scale stand disturbances). Yet, soil nutrient and water availability may also significantly affect recruitment, with seedlings of large-seed species often more tolerant of low nutrient and water availability, by virtue of their larger seed reserves and greater root growth rate (Augspurger 1984; Pons 1989; Burton and Bazzaz 1991; Leishman and Westoby 1994; Topoliantz and Ponge 2000).

Recruitment, particularly in small-seed tree species, can be further affected by litter accumulation, itself dependent on micro-topography and differences in litter production/persistence (Sydes and Grime 1981a, b; Beatty and Stone 1986; Peterson and Facelli 1992; Dzwonko and Gawronski 2002). For example, the primary root of seedlings of *Betula alleghaniensis* Britton cannot reach the mineral soil if seeds have germinated over a thick litter; however, if seeds have germinated under the litter, seedlings cannot emerge. Consequently, a litter-free substrate (e.g., small mounds) is necessary for seedling recruitment in this species (Linteau 1948; Hatcher 1966; Houle and Payette 1990; Houle 1992a). In contrast, seed germination and seedling emergence in large-seed tree species are not inhibited by a thick litter (Tubbs and Houston 1990); in fact, seeds of some species (e.g., *Quercus rubra* L.) may even benefit from being buried under thick litter by escaping from predators (Garcia and Houle 2005).

In this study, we identify the spatial structure of some resources and abiotic conditions significant for seedling emergence and survival, and determine how these environmental factors influence the recruitment of four species – *Fagus grandifolia* Ehrh., *Acer saccharum* Marsh., *Fraxinus americana* L., and *Ostrya virginiana* (Mill.) K. Koch – in a cold-deciduous forest of southern Québec, Canada (hereafter referred to as *Fagus*, *Acer*, *Fraxinus*, and *Ostrya*). These species are shade-tolerant hardwoods (Baker 1949), overlap in their distribution in eastern North America (Greller 1988), and typically maintain an abundant seedling bank (Houle 1991). However, because they differ in mean seed size (*Fagus*, 0.284 g; *Acer*, 0.074 g; *Fraxinus*, 0.045 g; and *Ostrya*, 0.015 g; Hewitt 1998), we might expect an increase from *Fagus*, through *Acer* and *Fraxinus*, to *Ostrya* in the control of environmental variables on seedling emergence and survival (Grime and Jeffrey 1965; Leishman and Westoby 1994). Compared to other studies done in controlled environments (e.g., Walters and Reich 2000), we consider natural patterns of co-variability in resources and conditions and their effects on seedling emergence and survival in a specific ecosystem.

Materials and methods

The study site is located on Île-aux-Grues (47°02′ N, 70°33′ W), an island of the Montmagny archipelago, approximately 70 km east of Quebec City (Québec, Canada). The central part of the island has been cleared for agriculture, but the periphery and the southwest section of the island are still forested. The dominant tree species in the southwest section of the island (old-growth forest), where the study was carried out, are *Fagus*, *Acer*, *Fraxinus*, and *Ostrya* (all typical of the deciduous forest of eastern North America). Mean annual temperature is 4.4 °C and annual precipitation totals 1087 mm of which 23% falls as snow

(Atmospheric Environment Service 1993). The frost-free period lasts *ca.* 146 days (Atmospheric Environment Service 1982) and there are 1685 degree-days above 5 °C.

A 50 m × 50 m area (0.25 ha) was delimited within a section of the forest in the fall of 1998 and subdivided into 100 5 m × 5 m quadrats. A 1 m × 1 m plot was marked in two diagonally opposite corners of every other quadrat, in a checkerboard pattern (for a total of 100 plots). On each plot, starting early May 1999, newly-emerged seedlings of *Fagus*, *Acer*, *Fraxinus*, and *Ostrya* were marked and followed every 2 weeks until the end of September. Seedling emergence values for 1999 were representative of average to good years for the area (Houle 1991, 1992a, b, 1994, 1995). Each tree with a diameter ≥5 cm at 125 cm from the soil surface was measured and mapped, and total basal area was calculated.

One soil sample (12.6 cm^2 × 10 cm; H horizon and top of A horizon) was taken at *ca.* 10 cm from each 1 m^2 plot on June 28, approximately 3 days after a rain event. Samples were passed through a 2-mm mesh size sieve to remove coarse debris, they were weighed and then dried at 75 °C for 24 h. Soil dry mass was determined and used to calculate substrate moisture content. Soil N concentration was then measured using the macro-Kjeldahl method. Total soil N can have an important control on N mineralization, standing pools, and plant availability, although other factors may affect N mineralization rates and standing mineral N pools (Vitousek 1985). Consequently, soil N concentration only represents a crude index of the N available to the tree seedlings. Photosynthetic photon flux density (PPFD) was measured at 25 cm from the soil surface in the center of each 1 m^2 plot on June 21, between 10:00 h and 12:00 h, under uniform cloudy conditions with a photometer (model 189, Li-Cor, Inc., Lincoln, NE, USA). Mean leaf litter thickness was calculated from five measurements made on each 1 m^2 plot (one measurement in each corner of the plot and one in the center) after leaf fall at the end of October: a pin was inserted vertically through the litter and the number of leaves intercepted was counted. Elevation was measured to the nearest 10 cm over the 0.25 ha area at 121 regularly-spaced points with a Pentax AL-M5C level and later standardized so that the lowest elevation on the surface corresponded to 0 m.

Statistical analyses

Moran's I

Spatial autocorrelation was determined for each variable using Moran's *I*. The equation for this index ($I_{(d)}$) is as follows (Legendre and Legendre 1998):

$$I_{(d)} = \frac{(1/W)\sum_{h=1}^{n}\sum_{i=1}^{n} W_{\mathrm{hi}}(y_h - \bar{y})(y_i - \bar{y})}{(1/n)\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

y_h and y_i represent the value of the variable observed at points *h* and *i*, respectively; w_{hi} represents a weight (1 if both points *h* and *i* are in the class interval (*d*) considered; 0 otherwise); *n* represents the total number of points; W is the sum of the weights w_{hi} for a given distance class, i.e., the number of pairs used to calculate $I_{(d)}$. Moran's *I* typically varies between −1 (repulsion) and +1 (contagion), with non significant values close to 0 (Legendre and Fortin 1989). Moran's *I* can be calculated for different distance classes (*d*) and presented in a correlogram. Before testing individual values of *I* for significance, the correlogram must be globally significant (i.e., one value must be significant at $p \leq 0.05/k$, with *k* representing the number of distance classes; Bonferroni criterion). We used 12 distance classes of *ca.* 5.8 m each for seedling emergence and survival, soil N and moisture, PPFD, leaf litter thickness and elevation, and 12 distance classes each of *ca.* 5.3 m for tree basal area. The coordinates of each 1 m^2 plot or 25 m^2 quadrat center was used for the analyses. We used distance classes of 5.8 and 5.3 m to optimize the number of pairs per distance category and because the distance between plot centroids within a quadrat was *ca.* 5.7 m.

Partial Mantel tests

Two variables may appear to be correlated because of their common dependence on a third variable, e.g., location in space. Thus, in the presence of spatial autocorrelation, it is necessary to remove this 'spatial' effect before concluding that two variables are indeed correlated (in a manner similar to a partial correlation test). This can be done with a partial Mantel test (Smouse et al. 1986). A significant coefficient of correlation for the partial Mantel test (*r*, with a value from −1 to +1) indicates that the relationship that exists between two variables is not related to a common spatial structure (Legendre and Fortin 1989).

Relationships between the spatial patterns of seedling emergence and survival, tree basal area, PPFD, soil N and moisture, leaf litter, and elevation were thus estimated with partial Mantel tests (Legendre and Fortin 1989). *P*-levels for these tests were determined by a permutation procedure (from 1000 permutations). Because each seedling data set (emergence and survival) was used repeatedly for the analyses with the five environmental variables, we used the more stringent alpha-level of 0.01 (Bonferroni correction). We used the coefficient of variation (CV = 100*SD/mean) as an estimate of heterogeneity (Kleb and Wilson 1997; Gallardo 2003; Lundholm and Larson 2003). Autocorrelation analyses and partial Mantel tests were performed with the R software package of Legendre and Vaudor (1991).

Results

Stand characteristics

Total basal area was 36.2 m^2/ha on the study area. *Fagus* (6.9 m^2/ha), *Fraxinus* (6.2 m^2/ha), and *Acer* (6.0 m^2/ha), were more or less equally abundant in terms of basal area, while *Ostrya* (1.7 m^2/ha), was much less important. Together these four species accounted for over 57% of the total basal area (Table 1). Among the other species present on the study area were *Betula papyrifera* Marsh., *Quercus rubra*, *B. alleghaniensis*, and *Acer rubrum* L. (Table 1). *Acer* was clearly the dominant species in terms of stem density. Spatial heterogeneity (CV) in basal area and in tree density were high for all species, but especially so for *Fraxinus* and *Ostrya* (Table 1).

The correlogram for *Acer* basal area indicated positive autocorrelation at *ca.* 40 and 65 m and negative autocorrelation at *ca.* 60 m (Figure 1). Autocorrelation at 60 and 65 m was based on too few points to be interpreted validly (Legendre and Fortin 1989). Positive autocorrelation at 40 m was due to a complex of patches (5–10 m in diameter) separated by approximately 40 m, in a matrix of low basal area values. The correlogram for *Fraxinus* (Figure 1) was globally significant (Bonferroni criterion), and suggested relatively small patch size (<5 m in diameter) and inter-patch distances of 10–15 m. The correlograms for *Fagus* and *Ostrya* were not globally significant (Figure 1), but visual inspection of their distribution suggested that *Ostrya* tended to occupy mostly the lower and right sections of the study area, while *Fagus* was concentrated in the left and upper left sections.

Seedling emergence

Seedling density was approximately three times higher at emergence in the spring, and ten times higher in the fall, for *Acer* and *Fagus* than for the other two species (Table 2 and Figure 2). Seedling density was highly heterogeneous both at emergence and in the fall, particularly for *Acer* (Table 2). However, increases in heterogeneity during the season were more pronounced in *Fraxinus* and *Ostrya*: indeed, CV-values almost doubled from spring to fall for these two species.

There was significant spatial autocorrelation for the seedling density of all four species at emergence (Figure 2). All correlograms showed positive values of Moran's *I* at small distances (up to the second or the third distance class, i.e., up to *ca.* 10 or 15 m) and significant negative values at greater distances (from the fifth, sixth, or the seventh distance class, i.e., from *ca.* 30, 35, or 40 m; Figure 2). However, specific patterns differed among correlograms. For *Acer*, *Fraxinus*, and *Ostrya*, patch size diameter corresponded to

Table 1. Tree density and basal area (with their respective coefficient of variation, CV) calculated from 100 25-m^2 quadrats at the study site at Île-aux-Grues (Québec, Canada)[a]. Mean ± SE.

Species	Density (trees/ha)	CV density (%)	Basal area (m^2/ha)	CV basal area (%)
Acer saccharum	436 ± 60	142	6.0 ± 1.5	260
Fagus grandifolia	152 ± 28	194	6.9 ± 1.7	242
Fraxinus americana	128 ± 28	217	6.2 ± 2.0	313
Ostrya virginiana	80 ± 28	236	1.7 ± 0.5	318

[a] Also present on the study area: *Abies balsamea* (L.) Mill., 0.06 m^2/ha; *Acer pensylvanicum* L., 0.01 m^2/ha; *Acer rubrum*, 1.56 m^2/ha; *Amelanchier arborea* (Michx. f.) Fern., 0.08 m^2/ha; *Betula alleghaniensis*, 2.10 m^2/ha; *Betula papyrifera*, 5.80 m^2/ha; *Quercus rubra*, 5.53 m^2/ha; *Tilia americana* L., 0.32 m^2/ha.

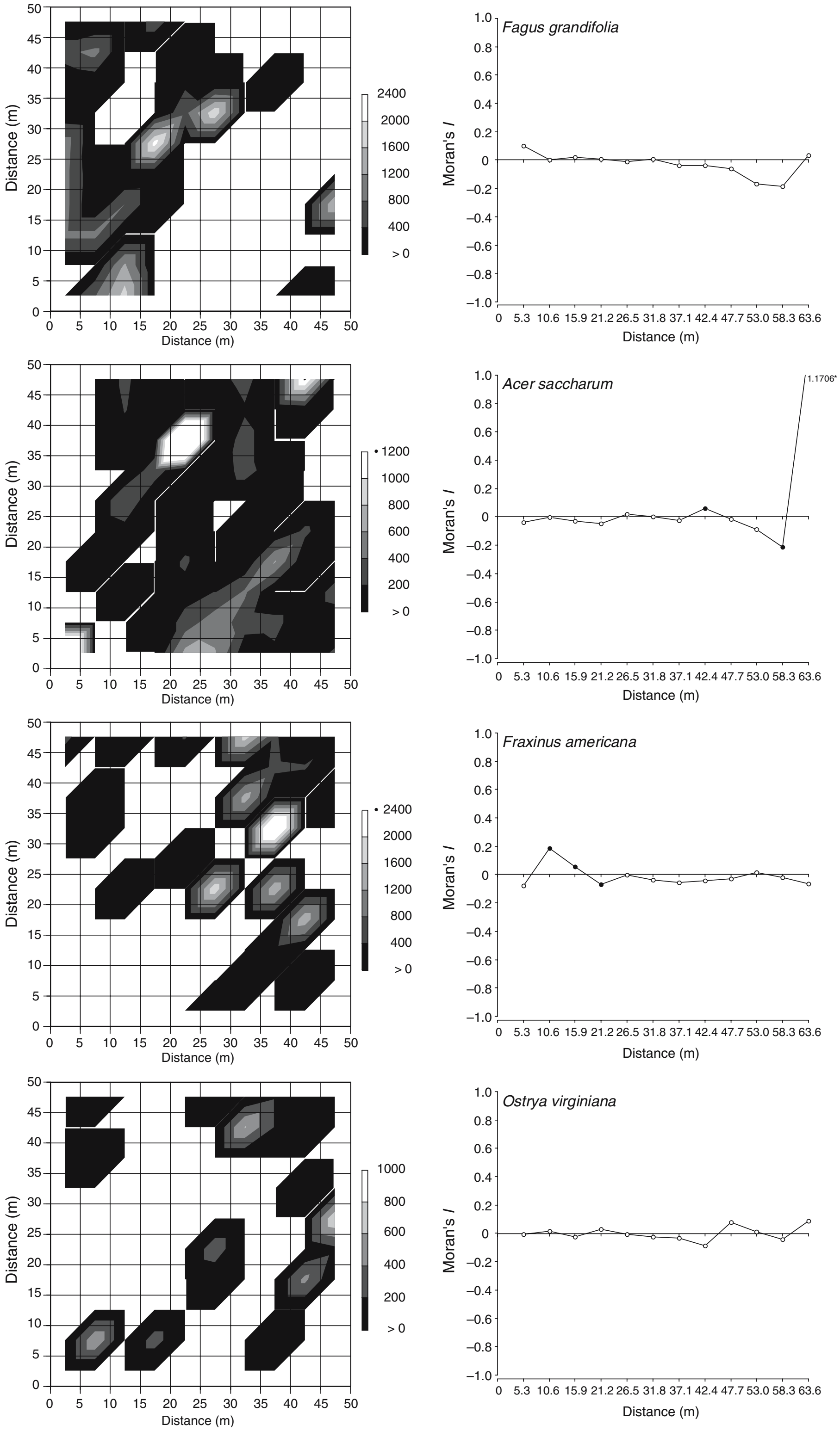

Figure 1. Left: Spatial pattern of basal area ($cm^2/25\ m^2$) at the study site (values increase from darker to lighter shades). Right: Matching spatial correlograms (significant Moran's I values are represented by black dots on globally significant correlogram, $p \leq 0.004$; Bonferroni criterion).

Table 2. Summary results of the demographic variables studied for *Fagus grandifolia*, *Acer saccharum*, *Fraxinus americana*, and *Ostrya virginiana* (species in decreasing order of seed size) at Île-aux-Grues, Québec. For seedling density and survival probability: mean ± SE.

Demographic stage/ process	Variables	Species			
		F. grandifolia	*A. saccharum*	*F. americana*	*O. virginiana*
Seedling at emergence (in May)	Density (#/m^2)	1.47 ± 0.34	1.50 ± 0.53	0.55 ± 0.10	0.45 ± 0.10
	Coefficient of variation (%)	234	350	176	233
	Autocorrelation	Yes	Yes	Yes	Yes
	Environmental variables to which emergence is correlated	Elevation[a]	PPFD[b]	Soil N, soil moisture	None
	Relationship between seedling at emergence and tree basal area	+	+	+	+
Seedling in the fall (in September)	Density (#/m^2)	0.70 ± 0.20	0.83 ± 0.37	0.09 ± 0.03	0.06 ± 0.02
	Coefficient of variation (%)	289	446	320	398
	Autocorrelation	Yes	Yes	Yes	Yes
	Relationship between seedling density at emergence and in fall	+	+	ns[c]	ns[c]
Seedling survival (from May to September)	Probability	0.40 ± 0.07	0.49 ± 0.07	0.22 ± 0.07	0.21 ± 0.08
	Coefficient of variation (%)	104	86	182	191
	Autocorrelation	No	No	No	No
	Environmental variables to which survival is correlated	None	None	None	PPFD[b]
	Relationship between seedling survival and density at emergence	+	+	ns[c]	ns[c]

[a] This relationship, determined by a partial Mantel test, gives a negative r, which indicates that large differences in seedling emergence are associated with small differences in elevation, or *vice versa*.
[b] PPFD: photosynthetic photon flux density.
[c] ns: not significant.

approximately 10 m, with an unstructured pattern between patches (Figure 2). For *Fagus*, emergence pattern approximated a wave, going from the left (crest of the wave) to the central part (low of the wave) of the 2500 m^2 area (Figure 2).

Seedling emergence was positively correlated to tree basal area (for the 25 m^2 quadrats) in *Fagus* (r = 0.298, p = 0.029) and *Acer* (r = 0.589, p = 0.017), but not in *Fraxinus* (r = –0.079, p = 0.237) and *Ostrya* (r = 0.012, p = 0.263).

Seedling survival

Seedling survival was approximately twice higher and less heterogeneous for *Fagus* and *Acer* than for *Fraxinus* and *Ostrya*. However, survival probability was not spatially structured for either one of the four species (data not shown). Survival was significantly and positively correlated to density at emergence for *Acer* (r = 0.194, p = 0.018) and *Fagus* (r = 0.423, p = 0.001), but not for *Fraxinus* or *Ostrya* (r = 0.068, p = 0.156 and r = 0.074, p = 0.126, respectively).

Environmental variables

Soil N averaged 0.88% (range: 0.39–1.99%) and soil moisture 39.4% (range: 23.8–64.0%; Table 3). Mean PPFD was 7.0 μmol/m^2/s (range: 1.2–20.9 μmol/m^2/s) and represented approximately 4% of full light (on the same day, we obtained an average reading of 178 μmol/m^2/s, in

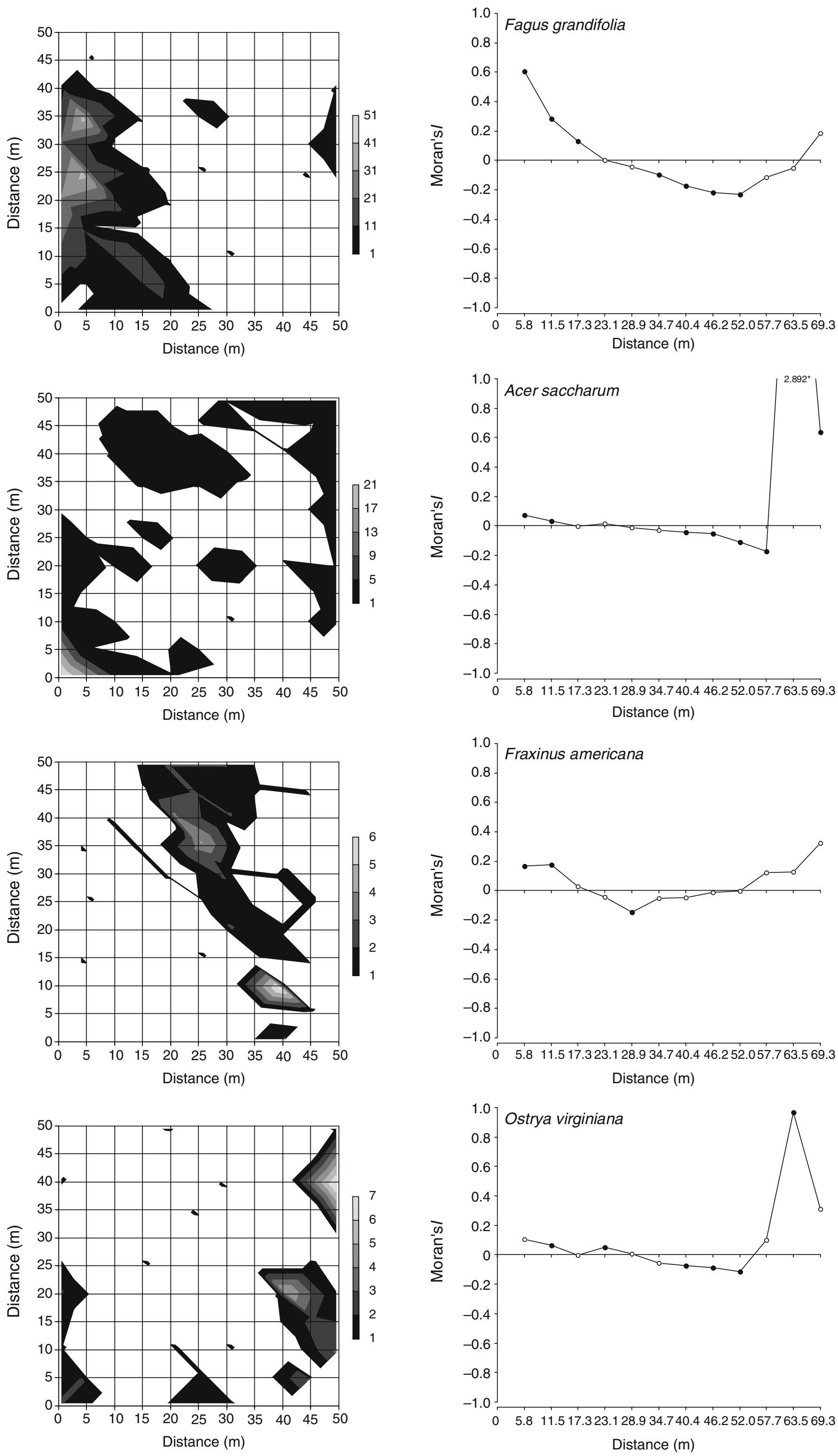

Figure 2. Left: Spatial pattern of seedling density (seedlings/m^2) at the study site (values increase from darker to lighter shades). Right: Matching spatial correlograms (significant Moran's *I* values are represented by black dots on globally significant correlogram, $p \leq 0.004$; Bonferroni criterion).

Table 3. Environmental variables measured at the study site at Île-aux-Grues (Québec, Canada) and their respective coefficient of variation (CV; $n = 100$).

Variable	Mean ± SE	CV (%)
Soil N (%)	0.88 ± 0.04	41
PPFD (μmol/m^2/s)	7.0 ± 0.5	67
Soil moisture (%)	39.4 ± 0.8	21
Leaf litter thickness (leaf layers)	2.9 ± 0.2	62
Elevation (m)	0.99 ± 0.06	57

a nearby open field). The difference between the minimum and maximum elevations was 2.1 m. Leaf litter thickness averaged 2.9 leaf layers (range: 0–10 leaf layers). Of the five environmental variables, PPFD was the most heterogeneous and soil moisture the least heterogeneous (Table 3). Soil N, elevation, and leaf litter thickness had intermediate CV values.

Soil N, PPFD, and leaf litter thickness all showed significant positive spatial autocorrelation at small distances (up to *ca*. 10 m, which grossly corresponds to patch size diameter; Figures 3, 4). However, at greater distances, the patterns were more complex, although soil N patterns appeared more coherent (three high-value patches 25–30 m apart). For elevation, values of Moran's I were positive up to *ca*. 15 m; then, they became negative up to 50 m and positive again (although non significantly so) at greater distances (Figure 4), suggesting 15–20 m diameter patches separated by *ca*. 35 m. There was no significant spatial autocorrelation for soil moisture (Figure 3). Pairwise comparisons between environmental variables revealed only one significant correlation: that between soil N and moisture ($r = 0.702$, $p = 0.001$).

Relationships between seedling emergence and survival and environmental variables

Fagus seedling emergence was associated with elevation ($r = -0.172$, $p = 0.001$): this negative r indicates that large differences in seedling emergence were associated with small differences in elevation, and *vice versa*. *Acer* seedling emergence was positively correlated to PPFD ($r = 0.285$, $p = 0.002$) and that of *Fraxinus* to soil N ($r = 0.284$, $p = 0.003$) and moisture ($r = 0.243$, $p = 0.004$). However, soil N and moisture were positively correlated (see above). When we removed the effect of soil N on the relationship between *Fraxinus* seedling emergence and soil moisture with a partial Mantel test, r decreased to 0.063, a value no longer significant ($p = 0.124$). Removing the effect of soil moisture on the relationship between *Fraxinus* seedling emergence and soil N (with a partial Mantel test), produced an $r = 0.163$ ($p = 0.011$), indicating that the relationship between *Fraxinus* emergence and soil moisture was essentially caused by a correlation between soil N and moisture.

Ostrya seedling survival was positively correlated to PPFD ($r = 0.201$, $p = 0.012$). None of the other correlations between seedling emergence and survival and the environmental variables were significant.

Discussion

Our initial prediction (increase in the control of environmental variables on seedling emergence and survival with a decrease in seed size) was only partially confirmed for the species studied. Indeed, seedling survival was lower and more heterogeneous in *Fraxinus* and *Ostrya*, the two small-seed species, and light availability (PPFD) was associated with the seedling survival of the smaller-seed, yet very shade tolerant *Ostrya* (Grime and Jeffrey 1965; Leishman and Westoby 1994; Walters and Reich 2000).

Environmental variables

Of the five environmental variables we studied (soil N and moisture, PPFD, leaf litter, and elevation), only soil moisture – the least heterogeneous variable – was not spatially structured. When present, autocorrelation was positive at small distances, indicating relatively fine-scale patterns of variation. Similar results have been reported for different types of systems, even forest ecosystems (Lechowicz and Bell 1991; Gross et al. 1995; Farley and Fitter 1999; Nicotra et al. 1999; Gallardo 2003), which traditionally have been considered relatively homogeneous. Disturbances such as those associated with tree death, windthrow, the activity of herbivores, etc. may cause

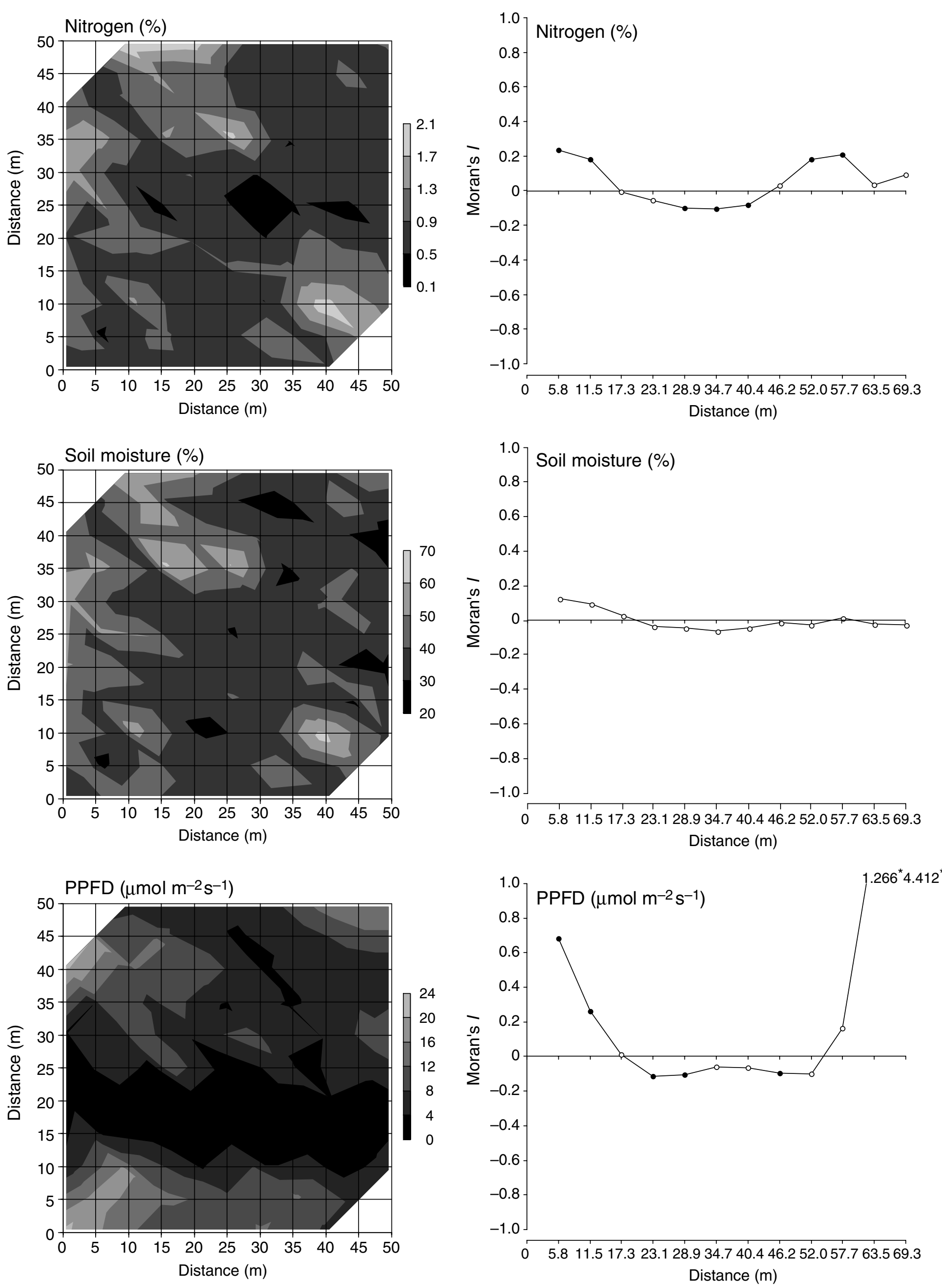

Figure 3. Left: Spatial pattern of soil nitrogen concentration (%), soil moisture content (%), and photosynthetic photon flux density (PPFD: μmol/m^2/s^1) at the study site (values increase from darker to lighter shades). Right: Matching spatial correlograms (significant Moran's *I* values are represented by black dots on globally significant correlogram, $p \leq 0.004$; Bonferroni criterion).

such fine-scale patterns in environmental variables, with significant effects on seedling recruitment (Hutnik 1952; Putz 1983; Houle 1992a).

While soil N and moisture were positively correlated, PPFD was not associated with either one of the two soil variables. Because soil resources and light are significant for seedling growth and survival in the forest understory, spatial discordance in resources as reported here (contrary to the concomitant increase of light and soil resources in forest canopy gaps of moderate size) may be of great importance for maintaining tree richness by

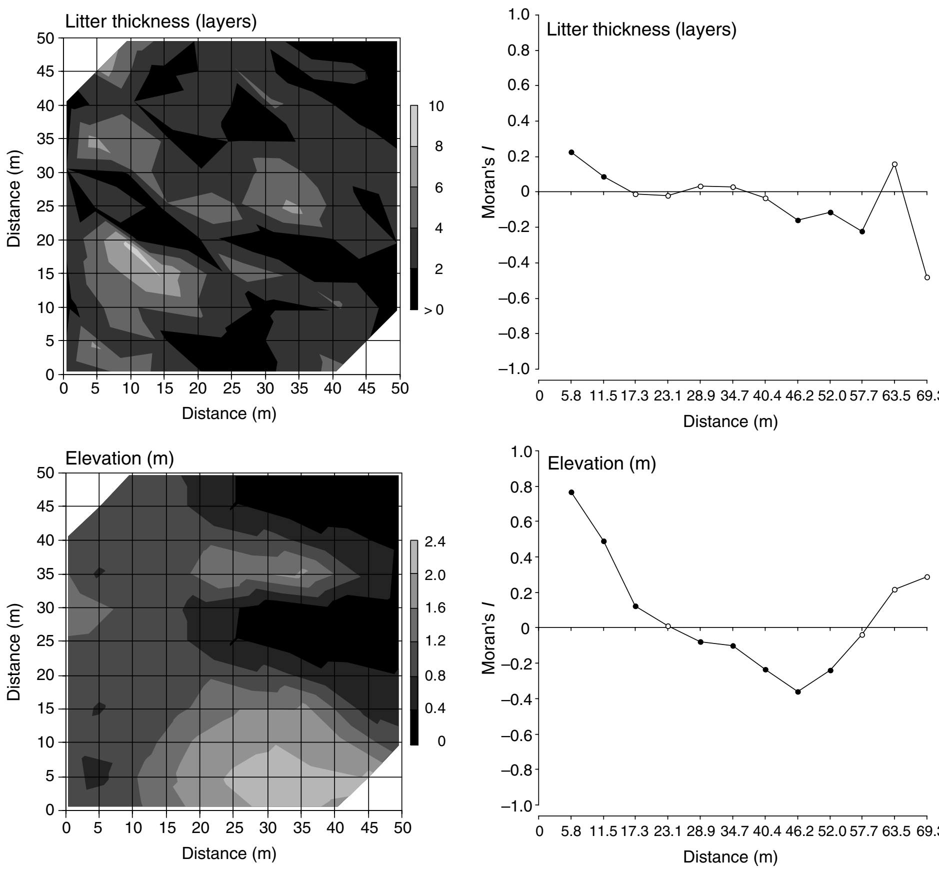

Figure 4. Left: Spatial pattern of litter thickness (layers) and elevation (m) at the study site (values increase from darker to lighter shades). Right: Matching spatial correlograms (significant Moran's I values are represented by black dots on globally significant correlogram, $p \leq 0.004$; Bonferroni criterion).

creating a diverse mosaic of conditions for recruitment (assuming niche specialization; Grubb 1977; Walters and Reich 1996, 2000; Nakashizuka 2001). Yet, heterogeneity in any one of the environmental variables was low to moderate in comparison to that in seedling density, PPFD being the most spatially heterogeneous of the environmental variables analyzed. However, because soil N, soil moisture, and PPFD were measured only once and because these variables are known to change over time, their long-term spatial patterns and their relationship with seedling emergence and survival must be discussed with great care.

Seedling emergence and survival

There was positive spatial autocorrelation and high heterogeneity in the seedling density (at emergence) of all four species studied. However, because survival was positively density-dependent in *Acer* and *Fagus* (higher survival in higher density patches) spatial heterogeneity increased through the season and initial autocorrelation patterns were reinforced (Hubbell 1980; McCanny and Cavers 1987; Houle 1995; but see: Janzen 1970). These results suggest that conditions that enhanced seedling emergence also had a positive effect on seedling survival. Assuming this process continues through time, it would tend to create a relatively stable patch structure. However, we may expect intraspecific competition to increase as seedlings become juveniles and, then, survival to become negatively density-dependent. Although heterogeneity in seedling density was also higher in the fall for *Fraxinus* and *Ostrya*, the immediate cause was different: indeed, it was related to highly heterogeneous survival patterns and resulted in a lack of relationship between spring and fall density patterns (Houle 1995). For these two species, assuming these processes continue through time, a shifting mosaic of recruitment would result with a highly unpredictable spatial dynamics.

Significant positive autocorrelation and high heterogeneity of seedling density at emergence may be explained by a number of interacting factors such as: tree dispersion pattern, fecundity of individual trees (which has been shown to be related to basal area; Greene and Johnson 1994), seed dispersal ability, the activity of seed predators, and heterogeneity of the environmental variables influencing seed germination and seedling emergence (Schupp and Fuentes 1995). Density at emergence was positively correlated with tree basal area in *Fagus* and *Acer*, emphasizing the influence of tree dispersion pattern and fecundity (associated with basal area) and limited seed dispersal on seedling recruitment patterns (despite apparent differences between species in dispersal ability: *Fagus*, zoochory; *Acer*, anemochory). In sharp contrast, density at emergence and tree basal area were not correlated in *Fraxinus* and *Ostrya* (both anemochores). By virtue of their small seed size, these two species may be considered to have a better primary dispersal ability than *Fagus* and *Acer*, which have larger seeds (Green 1983).

As the season progressed, heterogeneity in seedling density increased in all four species, but particularly so for *Fraxinus* and Ostrya. In *Ostrya*, survival was positively associated with PPFD, the most heterogeneous environmental variables ($r = 0.201$, $p = 0.012$). In the other three species, survival was independent of the environmental variables studied. Yet, heterogeneity in seedling density in the fall was still higher in *Acer* than in the other species (potentially the result of positive density-dependent survival and heterogeneous seedling emergence, see above).

Conclusion

If seed rain was saturating, seedling emergence should be as heterogeneous as the environmental variables that influence seed germination and seedling emergence and survival (Schupp and Fuentes 1995). In the present study, emergence was more heterogeneous than any of the environmental variables considered. Either seed rain was not saturating (because of restricted seed arrival), or factors other than those considered here influenced patterns of seedling establishment (e.g., predators, the effects of which may be spatially heterogeneous). Most likely, seed rain was not saturating, as has been shown for *Quercus rubra* on the same study site (Garcia and Houle 2005).

The processes described here may contribute to maintain tree species diversity. However, interannual variability in e.g., seed production, predator abundance, and micro-habitat suitability, also plays an important role by creating 'temporal' heterogeneity within the system. In addition, canopy gaps represent opportunities for species that require higher resource levels for successful recruitment (e.g., *Betula papyrifera*). The complex spatio-temporal mosaic that results from such time-space processes most likely is important for the maintenance of tree species diversity in this and other forest ecosystems.

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